

Al-Aqsa University-Gaza

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Data & System Security - Lab #1

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What is Kali Linux?

- Debian-based Linux distribution
- Specialized for penetration testing and digital forensics
- Contains over 600 pre-installed cybersecurity tools
- Used by ethical hackers and cybersecurity professionals

Kali Linux Desktop Overview

- Terminal - where you run commands
- File System - contains your files and folders
- Applications Menu - categorized tools for hacking and testing
- Browser - used for researching and downloading tools

Linux Terminal Basics

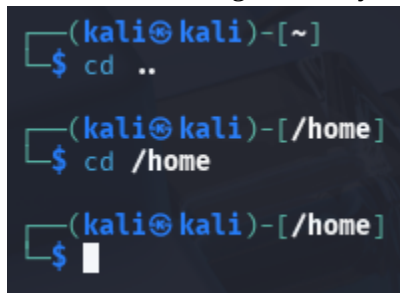
- pwd - Show current directory path

```
(kali㉿kali)-[~/Desktop]  
$ pwd  
/home/kali/Desktop
```

- ls - List files and folders

```
(kali㉿kali)-[~]  
$ ls  
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
```

- `cd <folder>` - Change directory



```
(kali㉿kali)-[~]  
$ cd ..  
  
(kali㉿kali)-[/home]  
$ cd /home  
  
(kali㉿kali)-[/home]  
$
```

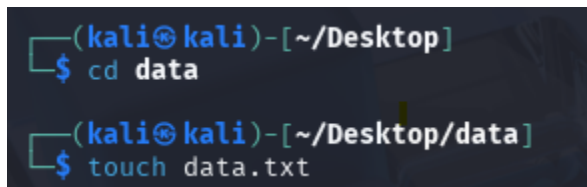
- `mkdir <folder>` - Create new folder



```
(kali㉿kali)-[~]  
$ cd Desktop  
  
(kali㉿kali)-[~/Desktop]  
$ mkdir data  
  
(kali㉿kali)-[~/Desktop]  
$
```

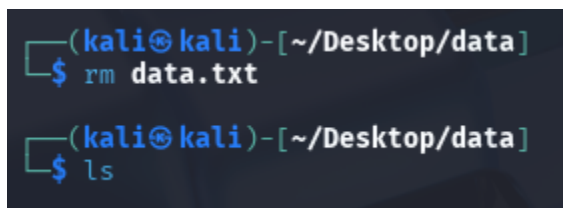
The image also shows a file explorer window with a folder named 'data' on the Desktop.

- `touch <file>` - Create new file



```
(kali㉿kali)-[~/Desktop]  
$ cd data  
  
(kali㉿kali)-[~/Desktop/data]  
$ touch data.txt
```

- `rm <file>` - Delete file



```
(kali㉿kali)-[~/Desktop/data]  
$ rm data.txt  
  
(kali㉿kali)-[~/Desktop/data]  
$ ls
```

- echo -to write in the text file

```
(kali㉿kali)-[~/Desktop/data]
$ touch data.txt

(kali㉿kali)-[~/Desktop/data]
$ echo "Hi Everyone" > data.txt
```

- cat -to Display Text file

```
(kali㉿kali)-[~/Desktop/data]
$ cat data.txt
Hi Everyone
```

- cp/mv - Copy or move files

```
(kali㉿kali)-[~/Desktop/data]
$ mkdir data2

(kali㉿kali)-[~/Desktop/data]
$ ls
data2  data.txt

(kali㉿kali)-[~/Desktop/data]
$ cp data.txt data2/

(kali㉿kali)-[~/Desktop/data]
$ cd data2

(kali㉿kali)-[~/Desktop/data/data2]
$ ls
data.txt
```

- clear - Clear the screen
clear

- rmdir <folder> - Delete folder

```
(kali㉿kali)-[~/Desktop]
$ mkdir sec

(kali㉿kali)-[~/Desktop]
$ rmdir sec
```

- whoami - Show current user

```
(kali㉿kali)-[~/Desktop/data/data2]
$ whoami
kali
```

- sudo - Run commands with admin rights
sudo (write your command)

Basic Network Commands

- ip a - Show IP address

```
(kali㉿kali)-[~/Desktop/data/data2]
$ ip -a
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }
       ip [ -force ] -batch filename
where  OBJECT := { address | addrlabel | fou | help | ila | ioam | l2tp | lin
k |
                        macsec | maddress | monitor | mptcp | mroute | mrule |
neighbor | neighbour | netconf | netns | nexthop | ntable
|
                        ntbl | route | rule | sr | stats | tap | tcpmetrics |
token | tunnel | tuntap | vrf | xfrm }
       OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
-h[uman-readable] | -iec | -j[son] | -p[retty] |
-f[amily] { inet | inet6 | mpls | bridge | link } |
-4 | -6 | -M | -B | -0 |
-l[oops] { maximum-addr-flush-attempts } | -echo | -br[ie
f] |
-o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filenam
e] |
-r[cvbuf] [size] | -n[etns] name | -N[umeric] | -a[ll] |
-c[olor]}
```

- ping 8.8.8.8 - Test internet connection

```
(kali㉿kali)-[~/Desktop]
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=93.1 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=308 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=79.3 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=735 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=128 time=362 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=128 time=742 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=128 time=94.3 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=128 time=79.4 ms
```

- ifconfig - Show network interfaces (might need installation)

```
(kali@kali)-[~/Desktop]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.59.137 netmask 255.255.255.0 broadcast 192.168.59.255
    inet6 fe80::7434:1527:1019:4531 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:78:59:3e txqueuelen 1000 (Ethernet)
    RX packets 196 bytes 18501 (18.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 256 bytes 24016 (23.4 KiB)
    TX errors 0 dropped 5 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 9 bytes 584 (584.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 9 bytes 584 (584.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Updating Kali System

- Use the following command:
- `sudo apt update && sudo apt upgrade`

Practical Exercise

1. Create a folder: `mkdir practice`
2. Navigate into it: `cd practice`
3. Create a file: `touch notes.txt`
4. Add text: `echo "Hello Kali" > notes.txt`
5. Display content: `cat notes.txt`